

The Impact of Music Packaging on Listener Utility
and Simulated Music Popularity

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1. Introduction

Music packaging is widely used in the music industry to attract listeners. However, whether packaging itself can change listener choice remains unclear. Packaging plays an important role not only in music industry, it also includes branding, social media, and product marketing. This project uses simulated model to investigate the effect of music packaging on listener choice.

2. Literature Review

2.1 Quality signalling and consumer utility

Caswell and Mojduszka (1996), drawing on **Hooker and Caswell (1996)**, food product quality attributes include food safety, nutritional value, packaging and process attributes. Consumers are willing to pay more for food products that provide additional safety because greater safety can increase consumer utility. However, based on **Swinbank (1993)**, **Caswell and Mojduszka** suggested that consumers are willing to pay less for each additional unit of safety, which indicated diminishing marginal utility from additional food safety. When information asymmetry exists between consumers and producers, consumers have difficulty distinguishing different quality of goods. Therefore, producers of high-quality goods have incentives to signal product quality to consumers, in order to get a price premium, distinguish from other goods (**Caswell and Mojduszka, 1996**). These suggest food labeling can serve as a quality signal for consumers to evaluate product under information asymmetry.

2.2 Consumer perception of labeling and quality signalling

Although food labels can serve as quality signals, their effectiveness depends on how consumers perceive and interpret the information presented. Consumers in order to keep away from health problem tend to avoid hazard choice on food products. Households will obtain and avoid the food products through different types of information beside them. For instance, if the food rehabilitation or impair health, the marginal attribute contribution to each food, or the actions that may reduce the marginal attribute contribution (**van Ravenswaay, 1995**). Consumers with different risk perceptions may have different purchasing decisions. If a consumer really cares about food safety, they are willing to purchase food with higher-than-optimal level of food safety (**Caswell and Mojduszka, 1996**). Therefore, different perception or understanding about the product requirements generates different decisions, next section we will apply these findings to music products.

2.3 Application to music product marketing

Music industry works differently to food industry, consumers can live without music, which means producers in order to sell recorded music, their challenge is to make consumers feel they need music. Artist branding is about clearly communicate with listeners who they are and what are they going to present to people to attract them (**Donham and Tammy, 2022**). Hence, artist branding contains information to listeners about the product content, listeners can choose them based on the information artists present on their brands. **Donham and Tammy** mentioned that “a strong brand increase a

product, company and artist's marketability and value." Caswell and Mojduszka (1996) provide a useful comparison from the food industry. They argue that food producers have incentives to supply higher product quality because of profitability or they have to. A similar mechanism may apply in music industry. Artists build their brand with clear information about their music and categories; it will differentiate one music product to another. Therefore, artists or producers may have incentives to invest in such signals when they expected increase their marketability and consumer valuation. However, these findings and reasoning developed above remain theoretical rather than empirically established. The following section builds a simple simulated model to examine this relationship, rather than modeling the full signalling process, this project isolates one simplified relationship: whether higher packaging intensity may increase listener utility at a diminishing marginal benefit.

3. Building Model

We assume diminishing marginal benefits of packaging intensity:

$$U = \log(P + 1) + \varepsilon$$

U is the utility function of each listener with 1 observable explanatory variable P, and one stochastic error term ε . P stands for the intense of the packaging, $P \in [0, \infty)$; $\varepsilon \sim N(0, 1)$, which included social influence, personal tastes, quality and other related factors.

$$\begin{aligned}\frac{dU}{dP} &= \frac{1}{(P + 1)} > 0 \\ \frac{d^2U}{dP^2} &= -\frac{1}{(P + 1)^2} < 0\end{aligned}$$

Therefore, the utility function is strictly concave and exhibits a diminishing marginal benefit of packaging intensity.

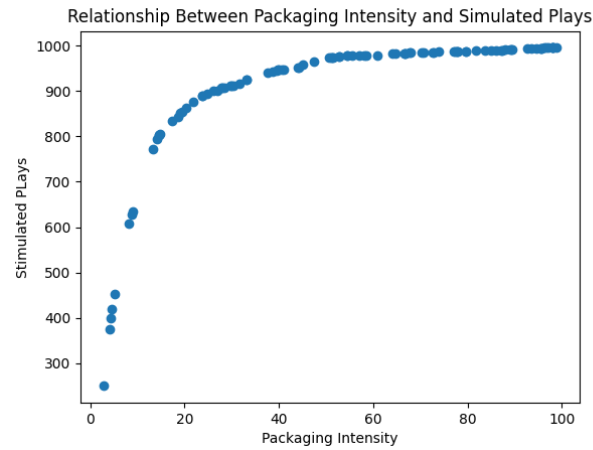
4. Methodology

The simulation generates 100 songs with uniformly distributed packaging intensity with range [0,100) and 1000 listener utility function, which $\varepsilon \sim N(0, 1)$. Each song was evaluated by one listener, repeat this process until 1000 listeners finished the evaluation. If the utility is greater than 2, listeners will choose to play the song, which reflected on scatter plot.

To further examine the relationship between packaging and utility, an additional analysis was conducted. Instead of simulated listener choices and play counts, I focused on plotting graphs of packaging intensity (P) against utility function (U); two scenarios were considered: $\varepsilon \sim N(0, 1)$ and $\varepsilon \sim N(0, 0.1^2)$, to illustrate the effect of random variation on the observed relationship.

5. Results

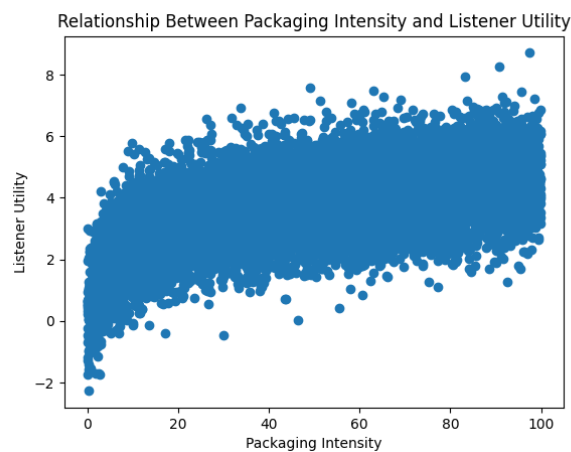
Figure 1 shows the relationship between packaging intensity and simulated plays. The results indicate that the marginal impact of packaging diminishes as packaging intensity increases. At low levels of packaging intensity, simulated plays increase rapidly. However, the growth slows as packaging intensity becomes stronger.



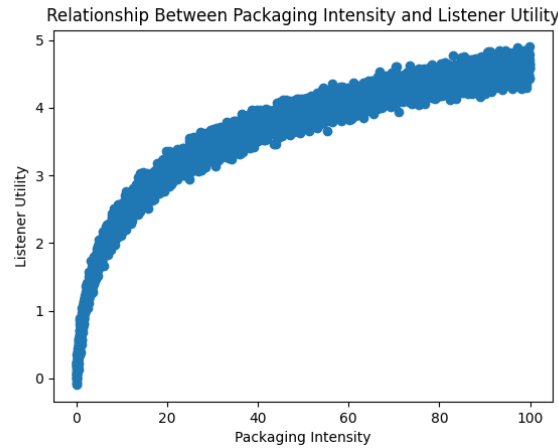
(Figure 1)

6. Additional Analysis

Additional analysis will focus on the relationship between music packaging and listener utility. In order to clearly illustrate the relationship, the large amount variation in Figure 2 obscures the relationship between packaging and utility, which has one standard deviation. Figure 3 shows the same relationship with 0.1 standard deviation. Therefore, due to smaller standard deviation, the relationship between packaging and utility becomes more apparent in Figure 3.



(Figure 2)



(Figure 3)

7. Results for Additional Analysis

According to Figure 3, the curve is similar to Figure 1, marginal benefit diminishes as packaging intensity increases, which can also illustrate the results in simulation model. However, the error term in real world cannot be this small, as Figure 2 shown, if we only focus on packaging, the random shock could disperse the impact.

8. Discussion

The results illustrated packaging appears to be more influential when initial packaging intensity is low. For music producer, packaging may be useful for attracting listeners, but it cannot guarantee unlimited growth. The impact of packaging may vary depending on external factors. Music producers should not only focus on packaging, but also consider external effects.

9. Limitations

The model of packaging and popularity does have many limitations, the impact of packaging is not absolutely independent; music quality, public tastes, artists reputation, and social influence could affect the results. According to additional analysis, with a large random shock, the impact of packaging will be dispersed. Therefore, this model is only adapted to a certain condition world, as other factors have little affect and independent to packaging.

Another limitation of this project is that the relationship between packaging intensity and listener choice are already determined by utility function. The logarithm in utility function described diminishing marginal benefits of packaging. However, different utility function may have different results, for instance linear or quadratic functions. Therefore, the results should be interpreted as a prediction in a specific situation rather than real-world phenomenon.

10. Conclusion

This project investigated the relationship between packaging and listener choice through a simplified simulated model. Additional analysis also suggested that the impact of packaging is sensitive to random shock.

The results suggested stronger packaging intensity increase simulated plays. However, the marginal effect diminishes as increases packaging intensity. For music producers, packaging might be useful to attract listeners, but not forever because they also have to consider other factors.

11. References

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